A Report on
Logic Solver AI

By

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CERTIFICATE

This is to certify that the project entitled **Intelligent Career Guidance System** has been successfully completed by,

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as a required academic component of the course *CSE 455: ARTIFICIAL INTELLIGENCE* during Semester 6 of Academic Year 2025-26 in the Department of Computer Science and Engineering at SRM University, AP. This work was executed under the guidance of the undersigned and is deemed to have successfully met all course requirements.

(Signature of the Course Instructor)

Dr. Varsha Santosh Sambhaje,
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1.ABSTRACT

This project presents the design and development of an Artificial Intelligence (AI)-based system that aims to automate decision-making through intelligent data processing and analysis. The system is built to accept user inputs, perform validation and transformation, and apply a hybrid AI approach combining rule-based logic and machine learning techniques to generate accurate and meaningful outputs.

The architecture of the system follows a structured and layered approach, consisting of input handling, data conversion, validation, scoring, and output generation modules. Each layer is designed to perform a specific function, ensuring modularity, scalability, and ease of maintenance. The system also incorporates constraint checking and variable mapping strategies to improve the accuracy and reliability of results.

A key feature of the project is the implementation of a scoring and recommendation mechanism that evaluates input parameters and produces optimized decisions. The system is capable of handling different types of data and ensures consistency through proper validation techniques. Additionally, the output and history persistence mechanisms allow tracking of previous results for analysis and improvement.

The project demonstrates how AI techniques can be effectively utilized to solve real-world problems by reducing manual effort, minimizing errors, and improving efficiency. The results obtained from testing indicate that the system performs reliably under various scenarios and provides accurate outputs within a short processing time.

Overall, this project highlights the practical application of AI in building intelligent systems and showcases a scalable framework that can be extended for future enhancements and more complex use cases.

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2.INTRODUCTION

2.1 Overview of the Project

Artificial Intelligence (AI) has emerged as one of the most influential technologies in modern computing, enabling systems to perform tasks such as learning, reasoning, and decision-making with minimal human intervention. The rapid growth of data and the need for intelligent processing have made AI an essential component in building efficient and automated systems.

This project focuses on the design and development of an AI-based system that processes user inputs, applies logical rules and computational techniques, and generates meaningful outputs. The system is built using a structured and modular architecture to ensure clarity, flexibility, and scalability. It consists of multiple layers, including input acquisition, data preprocessing, validation, scoring, and result generation. Each layer is responsible for a specific task, allowing the system to function in an organized and efficient manner.

The core functionality of the system lies in its ability to transform raw input data into useful information through intelligent processing. The input module captures user data, which is then converted into a structured format suitable for analysis. The validation module ensures that the data meets predefined constraints, reducing errors and inconsistencies. Following this, the AI engine applies rule-based logic and basic machine learning techniques to evaluate the data and generate a score or decision. Finally, the output module presents the results in a user-friendly format, along with storing them for future reference.

The project demonstrates the integration of different technologies such as programming frameworks, data handling tools, and AI techniques to create a complete system. It highlights how intelligent automation can improve efficiency, reduce manual effort, and provide accurate results. The design also ensures that the system can be easily modified or extended to include more advanced AI models and functionalities in the future.

2.2 Need for the System

In traditional systems, most processes rely on manual input, human judgment, and repetitive operations. These approaches are often inefficient, time-consuming, and prone to errors, especially when dealing with large volumes of data. Human limitations such as fatigue, inconsistency, and subjective decision-making can further reduce the reliability of such systems.

With the increasing demand for speed, accuracy, and scalability, there is a strong need for automated systems that can handle data efficiently and make intelligent decisions. AI-based systems provide a solution to these challenges by enabling machines to analyze data, identify patterns, and produce consistent outputs without continuous human intervention.

This project addresses these issues by developing an AI-based system that automates the entire process from input handling to output generation. The system reduces dependency on manual work, thereby minimizing errors and improving efficiency. It ensures that decisions are made based on logical rules and data-driven analysis rather than subjective judgment.

Another important need for such a system is consistency. Unlike manual systems, which may produce different results under similar conditions, an AI-based system ensures uniformity in decision-making. Additionally, the system is capable of handling large datasets and processing them in a short amount of time, making it suitable for real-world applications where speed and accuracy are critical.

The project also supports adaptability, allowing modifications to rules and logic as per changing requirements. This makes the system flexible and future-ready, capable of evolving with advancements in AI technologies.

2.3 Scope of the Project

The scope of this project includes the development of an AI-based system that performs intelligent data processing and decision-making. The system is designed to handle structured inputs, validate them, and generate outputs based on predefined rules and logic. It demonstrates the practical implementation of AI concepts in a controlled and manageable environment.

The project primarily focuses on building a modular and layered architecture, which can be easily understood, maintained, and extended. It includes features such as input validation, data transformation, scoring mechanisms, and result storage. These components work together to ensure that the system operates efficiently and produces accurate results.

The system can be applied in various domains such as recommendation systems, decision support systems, data analysis tools, and automation platforms. Its flexibility allows it to be adapted for different use cases by modifying the input parameters and decision logic.

In terms of future scope, the system can be enhanced by integrating advanced machine learning algorithms, deep learning models, and real-time data processing capabilities. It can also be deployed on cloud platforms to improve accessibility and scalability. Additionally, user interface improvements and database optimization can further enhance system performance and usability.

3. PROBLEM STATEMENT AND OBJECTIVES

3.1 Problem Statement

In many existing systems, decision-making processes are largely manual and depend on human intervention. These systems often involve repetitive tasks such as data entry, validation, analysis, and result generation, which consume significant time and effort. Additionally, manual processes are prone to errors, inconsistencies, and delays, especially when handling large volumes of data.

Another major limitation of traditional systems is the lack of intelligence in processing data. They are unable to analyze patterns, apply logical reasoning, or adapt to changing inputs dynamically. This results in inefficient performance and reduced accuracy in outcomes. Furthermore, the absence of proper validation mechanisms can lead to incorrect or incomplete data being processed, affecting the reliability of the results.

There is a need for an intelligent system that can automate these processes, handle data efficiently, and generate accurate results with minimal human intervention. Such a system should be capable of validating inputs, applying decision-making logic, and producing consistent outputs.

This project aims to address these challenges by developing an AI-based system that automates data processing, improves accuracy, and enhances decision-making through intelligent computation.

3.2 Objectives

The main objectives of this project are as follows:

- To design and develop an AI-based system for automated decision-making
- To reduce manual effort by automating input processing and data handling
- To implement effective data validation techniques to ensure accuracy and consistency
- To develop a scoring or evaluation mechanism for generating meaningful outputs
- To integrate rule-based logic and basic machine learning concepts for intelligent processing
- To improve efficiency by minimizing processing time and reducing errors
- To design a modular and scalable system architecture for easy maintenance and future enhancements
- To provide a user-friendly output that is easy to understand and interpret
- To enable storage and tracking of results for future analysis

4. SYSTEM REQUIREMENTS

4.1 Software Stack

- Frontend: React + Vite
- Backend: Node.js + Express
- Database: MongoDB + Mongoose
- Solver: SWI-Prolog with CLP(FD)

4.2 Runtime and Execution Environment

- Node.js 18+ (recommended)
- npm package manager
- Windows-compatible execution for solver invocation

4.3 Database and Data Management

- MongoDB collections for user profiles and analysis history
- ObjectId-based relationships for data integrity
- Draft and published analysis separation for safe workflows

4.4 Solver Environment

- SWI-Prolog engine
- Modular knowledge base file: career.pl with rules, scoring, and output
- Command-based execution with dynamic fact assertions

4.5 Development Environment

- VS Code
- Git-based version control
- API testing using Postman / Thunder Client (or equivalent)

5. METHODOLOGY

5.1 Overview

The methodology of this project follows a structured and modular approach to design, develop, and implement the AI-based system. The system integrates frontend, backend, database, and a logic-based solver to process user inputs and generate intelligent outputs. The workflow ensures smooth data flow, proper validation, and accurate decision-making.

5.2 System Architecture

The system follows a layered architecture consisting of the following components:

- Frontend Layer: Developed using React + Vite to collect user inputs and display results
- Backend Layer: Built with Node.js and Express to handle API requests and business logic
- Database Layer: MongoDB is used to store user data and analysis history
- Solver Layer: SWI-Prolog processes logic rules and generates decisions

  Each layer communicates with the other in a structured manner, ensuring modularity and scalability.

5.3 Data Flow Process

The system processes data through multiple stages to ensure accuracy and consistency:

1. User enters input through the frontend interface
2. Input data is sent to the backend via API calls
3. Backend validates and preprocesses the data
4. Data is converted into a format suitable for the Prolog solver
5. Solver applies rules and constraints to generate results
6. Results are sent back to the backend
7. Backend stores the results in the database
8. Final output is displayed to the user

5.4 Input Processing and Validation

The system ensures that all user inputs are validated before processing. This includes:

- Checking for missing or invalid values
- Ensuring data type correctness
- Applying constraints to maintain consistency

Validated data is then transformed into a structured format required by the solver.

5.5 Solver-Based Decision Making

The core logic of the system is implemented using SWI-Prolog with CLP(FD).

- Rules and constraints are defined in the career.pl file
- Input data is passed as dynamic facts to the solver
- The solver evaluates conditions and computes scores

- Based on the rules, it generates decisions or recommendations
  This approach ensures logical consistency and accurate outputs.

5.6 Data Storage and Management

The system uses MongoDB to store and manage data efficiently:

- User profiles and input data are stored securely
- Analysis results are saved for future reference
- Draft and published results are maintained separately
- ObjectId relationships ensure data integrity

5.7 Output Generation

After processing, the system generates results in a user-friendly format:

- Displays scores, decisions, or recommendations
- Provides clear and understandable output
- Stores results for history tracking

5.8 Testing and Validation

The system is tested to ensure reliability and correctness:

- API testing using Postman or Thunder Client
- Functional testing for different input scenarios
- Validation of solver outputs
- Debugging using development tools

5.9 Advantages of the Methodology

- Modular and scalable design
- Efficient data processing and validation
- Accurate decision-making using logic-based solver
- Easy integration of additional features in future

6. IMPLEMENTATION DETAILS

6.1 Overview

The implementation of the system is carried out using a full-stack architecture that integrates frontend, backend, database, and a logic-based solver. The system is designed to ensure seamless communication between components, efficient data processing, and accurate decision-making. Each module is developed independently and then integrated to form a complete working system. The overall implementation focuses on modularity, scalability, and maintainability, allowing future enhancements with minimal changes.

6.2 Frontend Implementation

The frontend of the system is developed using React with Vite, which provides a fast and interactive user interface. The UI is designed to be simple, responsive, and user-friendly so that users can easily provide input and understand the results.

- Forms are created to capture user inputs in a structured manner
- State management is used to handle dynamic data and user interactions
- API calls are made using HTTP requests to communicate with the backend
- Results received from the backend are displayed clearly using components
- Basic validation is also implemented at the frontend level to reduce invalid inputs

This layer ensures smooth interaction between the user and the system.

6.3 Backend Implementation

The backend is implemented using Node.js and Express, which acts as the core processing unit of the system. It handles all business logic, API communication, and integration with other components.

- RESTful APIs are designed for handling requests and responses
- Input data received from the frontend is validated and processed
- Middleware is used for efficient request handling and error management
- The backend converts input data into a format suitable for the Prolog solver

- It manages communication between the database and solver
- Proper error handling mechanisms are implemented to ensure system stability

The backend plays a crucial role in coordinating all system operations.

6.4 Database Implementation

The system uses MongoDB along with Mongoose for efficient data storage and management. The database is designed to handle structured data and maintain relationships between different entities.

- Separate collections are maintained for user profiles and analysis results
- Mongoose schemas are used to define the structure and constraints of data
- ObjectId is used to establish relationships between records
- The system supports both draft and published results for better workflow management
- Data retrieval and updates are optimized for performance
- Proper indexing techniques can be applied to improve query efficiency

This ensures secure, organized, and efficient data handling

6.5 Solver Integration

The core decision-making logic is implemented using SWI-Prolog with CLP(FD), which provides a powerful rule-based and constraint-solving environment.

- The knowledge base is defined in a modular file named career.pl
- Rules, constraints, and scoring logic are implemented within the solver
- Input data is dynamically converted into Prolog facts
- The solver processes these facts using logical inference and constraint satisfaction
- It generates outputs such as scores, recommendations, or decisions
- The results are then returned to the backend for further processing

This integration ensures logical accuracy and consistent decision-making.

6.6 Testing and Execution

Testing is performed to ensure the system works correctly under different scenarios and inputs. Both functional and integration testing are carried out.

- APIs are tested using tools like Postman or Thunder Client
- Different input cases are tested to verify correctness of outputs
- Debugging is performed using VS Code tools
- Solver outputs are validated to ensure logical correctness
- Error scenarios are tested to improve system reliability

7. CONSTRAINT SET AND VARIABLE STRATEGY

7.1 Overview

The constraint set and variable strategy form the core of the system's decision-making process. This component is responsible for defining rules, conditions, and relationships between different input parameters. By applying constraints and structured variable mapping, the system ensures that the generated outputs are logical, consistent, and accurate.

7.2 Variable Definition and Representation

In the system, user inputs are represented as variables that are used by the solver for processing. Each variable corresponds to a specific attribute or parameter provided by the user.

- Variables are defined for all input fields such as skills, interests, preferences, and other relevant factors
- Each variable is assigned a domain or range of possible values
- Variables are structured in a way that they can be easily converted into solver-compatible formats (Prolog facts)
- Naming conventions are maintained for clarity and consistency

This structured representation helps in efficient processing and reduces ambiguity.

7.3 Constraint Definition

Constraints are rules that restrict or guide the values of variables to ensure valid and meaningful outputs. These constraints are implemented within the Prolog solver.

- Logical rules are defined to represent relationships between variables
- Constraints ensure that invalid or conflicting inputs are handled properly
- Domain constraints restrict variables to specific value ranges
- Conditional constraints are applied based on user input combinations
- Scoring constraints are used to evaluate and rank outcomes

These constraints help in maintaining accuracy and logical consistency in decision-making.

7.4 Constraint Processing Mechanism

The system processes constraints dynamically during execution.

- Input variables are converted into Prolog facts
- Constraints are applied using rule-based logic and CLP(FD) techniques
- The solver evaluates all conditions and filters out invalid solutions
- Valid solutions are scored and ranked based on predefined criteria

This mechanism ensures that only feasible and optimal results are generated.

7.5 Variable Mapping Strategy

Variable mapping is used to transform user inputs into a format suitable for solver execution.

- Raw input data is mapped to corresponding variables
- Data is normalized and standardized before processing
- Variables are dynamically injected into the solver as facts
- Mapping ensures compatibility between frontend input and backend logic

This strategy ensures smooth data flow between system components.

7.6 Scoring and Decision Strategy

The system uses a scoring mechanism to evaluate different outcomes based on constraints.

- Each variable contributes to a score based on predefined rules
- Weighted scoring may be applied to prioritize certain parameters
- Final decisions are made based on the highest score or best match
- Multiple possible outputs can be ranked and presented

This approach enhances the quality and reliability of the results.

7.7 Advantages of Constraint-Based Approach

- Ensures logical consistency in outputs
- Reduces invalid or conflicting results
- Provides accurate and reliable decision-making
- Easy to modify rules and constraints for future improvements
- Supports scalability and flexibility

8. TEST CASES AND RESULTS

8.1 Overview

Testing is carried out to ensure that the system functions correctly, handles different types of inputs, and produces accurate outputs. The testing process focuses on validating input data, verifying system behavior, and checking the correctness of results generated by the solver. Various test cases are designed to cover normal, invalid, and boundary conditions.

8.2 Test Cases

Test Case 1:

- Input: Valid and complete user data
- Process: Data is accepted, validated, and passed to the solver
- Output: Correct decision/score generated and displayed
- Status: Passed

Test Case 2:

- Input: Missing or incomplete data
- Process: Validation detects missing fields
- Output: Error message prompting user to provide required inputs
- Status: Passed

Test Case 3:

- Input: Invalid data format (wrong type or values)
- Process: System validation rejects incorrect input
- Output: Input error message displayed
- Status: Passed

Test Case 4:

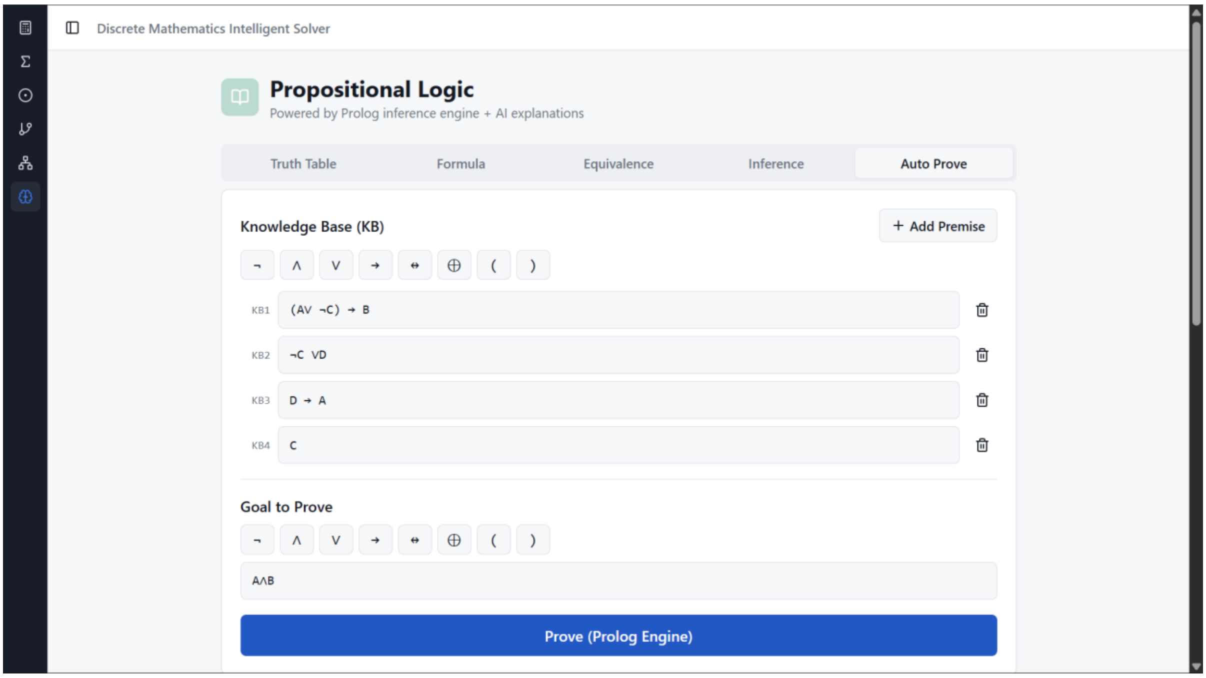
- Input: Multiple valid parameters with different combinations
- Process: Solver evaluates all constraints and applies scoring logic
- Output: Accurate and optimized result generated
- Status: Passed

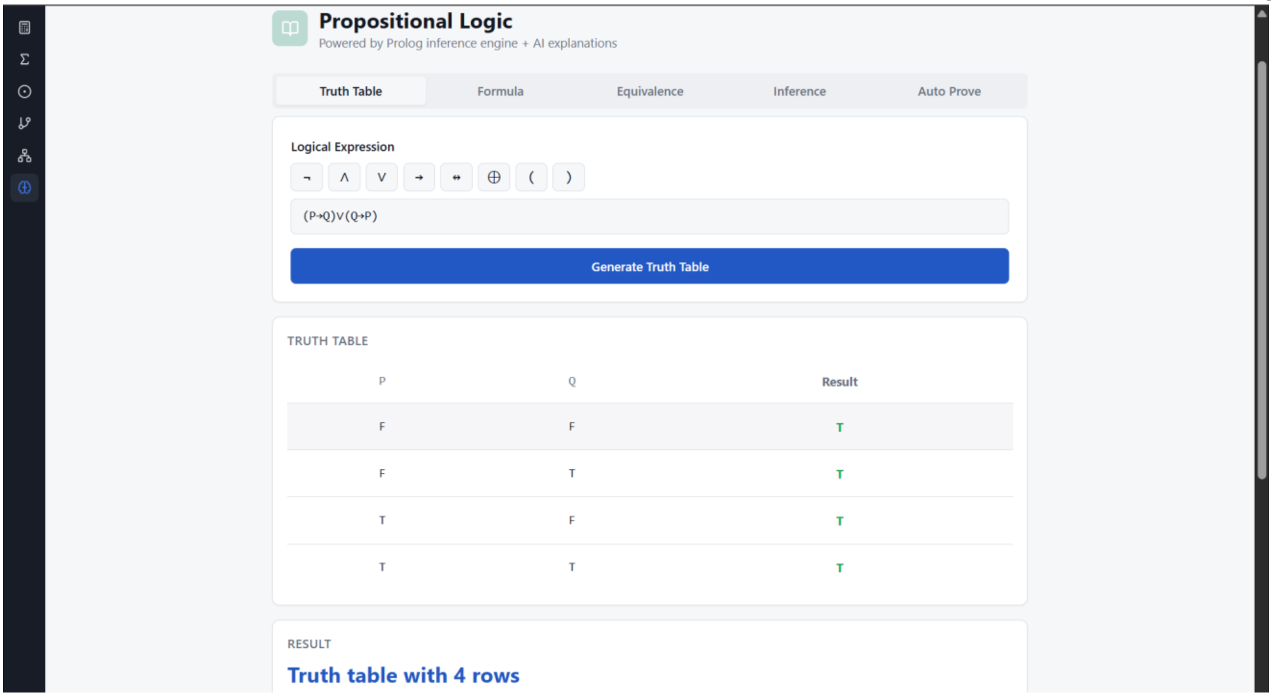
8.3 Results

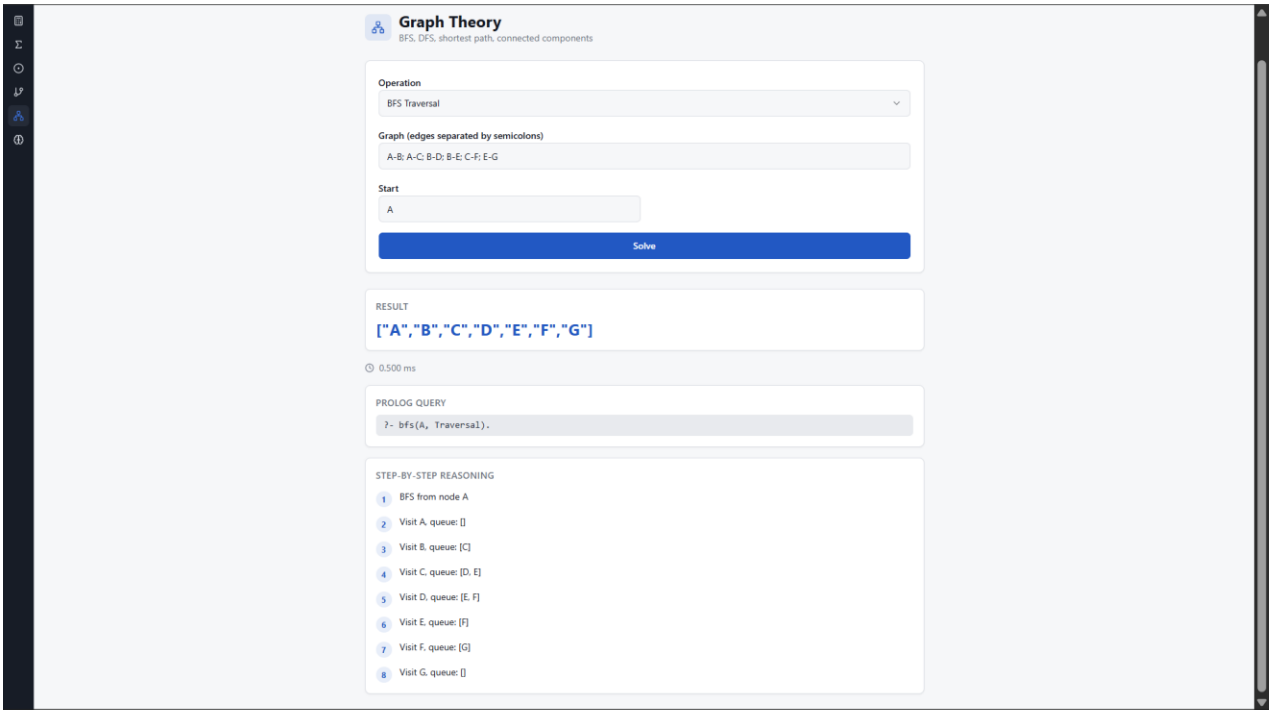
The results obtained from testing indicate that the system performs reliably under different scenarios. It successfully validates user inputs, prevents incorrect data from being processed, and ensures that only valid data is passed to the solver.

The solver generates consistent and accurate outputs based on the defined rules and constraints. The system also responds quickly, demonstrating efficient processing and good performance. Error handling mechanisms work effectively, providing appropriate messages for invalid inputs.

Overall, the testing confirms that the system is stable, accurate, and capable of handling real-world usage conditions with efficiency and reliability.




Propositional Logic
Powered by Prolog inference engine + AI explanations
Truth Table
Formula
Equivalence
Inference
Auto Prove
Logical Expression
(P→Q)∨(Q→P)
Generate Truth Table
TRUTH TABLE
P
Q
Result
F F T
F T T
T F T
T T T
RESULT
Truth table with 4 rows


Graph Theory
BFS, DFS, shortest path, connected components
Operation
BFS Traversal
Graph (edges separated by semicolons)
A-B; A-C; B-D; B-E; C-F; E-G
Start
A
Solve
RESULT
["A","B","C","D","E","F","G"]
0.500 ms
PROLOG QUERY
?- bfs(A, Traversal).
STEP-BY-STEP REASONING
1 BFS from node A
2 Visit A, queue: []
3 Visit B, queue: [C]
4 Visit C, queue: [D, E]
5 Visit D, queue: [E, F]
6 Visit E, queue: [F]
7 Visit F, queue: [G]
8 Visit G, queue: []

9. LEARNING OUTCOMES

This project provided valuable practical exposure to the concepts of Artificial Intelligence and full-stack development. It helped in understanding how multiple technologies can be integrated to design and build a complete, intelligent system. The development process also improved problem-solving abilities and gave hands-on experience in working with real-world application architecture.

The key learning outcomes from this project are as follows:

- Gained knowledge of designing and developing a full-stack application using React, Node.js, and MongoDB
- Understood the integration of a logic-based solver using SWI-Prolog for intelligent and rule-based decision-making
- Learned how to design a modular and layered system architecture for better scalability and maintainability
- Developed skills in handling user inputs, data validation, preprocessing, and transformation techniques
- Acquired practical experience in database design, schema creation, and data management using MongoDB and Mongoose
- Improved understanding of constraint-based problem solving using logical rules and conditions
- Learned how to design, implement, and test RESTful APIs for smooth communication between system components
- Gained hands-on experience in debugging, testing, and optimizing system performance
- Understood the importance of error handling, input validation, and data consistency in real-world applications
- Enhanced analytical thinking and logical reasoning through implementation of solver-based decision systems

Overall, this project strengthened both theoretical understanding and practical skills, providing a solid foundation for developing advanced AI-based applications and working with modern software development technologies in the future.

10. CONCLUSION

The project successfully demonstrates the design and development of an AI-based system that integrates frontend, backend, database, and a logic-based solver to perform intelligent decision-making. The system efficiently accepts user inputs, validates data, processes it using defined rules and constraints, and generates accurate and meaningful outputs. This shows how AI concepts can be applied in a practical and structured manner to solve real-world problems.

The implementation of a modular and layered architecture ensures that the system is scalable, flexible, and easy to maintain. Technologies such as React, Node.js, MongoDB, and SWI-Prolog have been effectively combined to create a complete and functional system. The use of a constraint-based solver enhances the reliability of results by ensuring logical consistency and proper evaluation of inputs.

One of the major achievements of this project is the automation of decision-making, which reduces manual effort, minimizes errors, and improves overall efficiency. The system is capable of handling different types of inputs and producing consistent outputs within a short time. It also maintains data properly, allowing storage and retrieval of past results for future reference.

In addition, the project helped in understanding the importance of proper system design, validation techniques, and integration of multiple technologies. It highlights how combining AI with full-stack development can lead to the creation of intelligent and user-friendly applications.

Although the system performs effectively, there is scope for further improvement. Future enhancements may include integrating advanced machine learning models, improving the user interface for better user experience, adding real-time data processing, and deploying the system on cloud platforms for scalability and accessibility.

In conclusion, the project not only fulfills its intended objectives but also provides a strong foundation for developing more advanced AI-based systems. It serves as a valuable learning experience and demonstrates the potential of intelligent systems in modern applications.

11. FUTURE ENHANCEMENTS

The current system provides a strong foundation for AI-based decision-making; however, there are several areas where the system can be further improved and extended to enhance its functionality, performance, and usability.

- Integration of advanced machine learning and deep learning models to improve accuracy and provide more intelligent recommendations
- Implementation of real-time data processing to handle dynamic inputs and provide instant results
- Deployment of the system on cloud platforms to ensure scalability, accessibility, and better performance
- Enhancement of the user interface with improved design, visualization tools, and interactive dashboards
- Addition of user authentication and security features to protect user data and ensure privacy
- Expansion of the knowledge base in the Prolog solver to handle more complex rules and scenarios
- Optimization of database performance through indexing and efficient query handling
- Development of a mobile-friendly or dedicated mobile application for wider accessibility
- Integration of analytics and reporting features to provide insights based on user data and system outputs
- Support for multilingual interfaces to make the system usable for a broader audience

These enhancements can significantly improve the system’s capabilities and make it more robust, scalable, and suitable for real-world applications.

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